Answer these Questions using CTE

1.Find the highest salaried employee of finance department.

Query :

```
WITH finance_dept AS (
   SELECT dept_no
   FROM departments
   WHERE dept_name = 'Finance'
),
finance_emps AS (
   SELECT emp_no
   FROM dept_emp
   WHERE dept_no IN (SELECT dept_no FROM finance_dept)
     AND to_date = '9999-01-01'
),
finance_salaries AS (
   SELECT emp_no, salary
   FROM salaries
   WHERE emp_no IN (SELECT emp_no FROM finance_emps)
     AND to_date = '9999-01-01'
)
SELECT emp_no, salary
FROM finance_salaries
WHERE salary = (SELECT MAX(salary) FROM finance_salaries);
```

Output:

	emp_no	salary
▶	28124	134662

2.Current manager of the production department.

Query:

```
WITH prod_dept AS (
   SELECT dept_no
   FROM departments
   WHERE dept_name = 'Production'
)
SELECT emp_no
FROM dept_manager
WHERE dept_no IN (SELECT dept_no FROM prod_dept)
  AND to_date = '9999-01-01';
```

Output:

	emp_no
▶	110420

3.How many managers are worked in the Human Resources department.

Query:

```
WITH hr_dept AS (
   SELECT dept_no
   FROM departments
   WHERE dept_name = 'Human Resources'
)
SELECT COUNT(DISTINCT emp_no) AS manager_count
FROM dept_manager
WHERE dept_no IN (SELECT dept_no FROM hr_dept);
```

Output:

	manager_count
▶	2

4.Salary of the Current Manager of the Production Department.

Query:

```
WITH prod_manager AS (
   SELECT emp_no
   FROM dept_manager
   WHERE dept_no = (
      SELECT dept_no FROM departments WHERE dept_name = 'Production'
   )
   AND to_date = '9999-01-01'
)
SELECT salary
FROM salaries
WHERE emp_no IN (SELECT emp_no FROM prod_manager)
  AND to_date = '9999-01-01';
```

Output:

	salary
▶	56654

5.Find the people whose name starts with a and end with s with only 5 letters , and their salaries.

Query:

```
WITH name_filter AS (
   SELECT emp_no, first_name
   FROM employees
   WHERE first_name LIKE 'a___s'
)
SELECT nf.first_name, s.salary
FROM name_filter nf
JOIN salaries s
ON nf.emp_no = s.emp_no
WHERE s.to_date = '9999-01-01';
```

Output:

first_name	salary
Aloys	50318
Alois	53157
Angus	57924
Aloys	95058
Angus	44773
Alois	71353
Angus	56868
Angus	77098

Note: The output is very long, so I have shown only a few sample records here. The remaining output is also available.

6.Find the employees whose hire date and From date are different ,how much difference is there for each one.

Query:

```
WITH emp_dates AS (
    SELECT e.emp_no, e.hire_date, s.from_date
    FROM employees e
    JOIN salaries s
    ON e.emp_no = s.emp_no
    WHERE s.to_date = '9999-01-01'
)
SELECT
    emp_no,
    hire_date,
    from_date,
    DATEDIFF(from_date, hire_date) AS date_difference
FROM emp_dates
WHERE hire_date <> from_date;
```

Output:

emp_no	hire_date	from_date	date_difference
10001	1986-06-26	2002-06-22	5840
10002	1985-11-21	2001-08-02	5733
10003	1986-08-28	2001-12-01	5574
10004	1986-12-01	2001-11-27	5475
10005	1989-09-12	2001-09-09	4380
10006	1989-06-02	2001-08-02	4444
10007	1989-02-10	2002-02-07	4745
10009	1985-02-18	2002-02-14	6205
10010	1989-08-24	2001-11-23	4474
10012	1992-12-18	2001-12-16	3285
10013	1985-10-20	2001-10-16	5840
10014	1987-03-11	2001-12-27	5405

Note: The output is very long, so I have shown only a few sample records here. The remaining output is also available.

7.Highest salaried manager and for which department he is present in, i need dept name ,and manager emp_no as output

Query:

```
WITH current_managers AS (
   SELECT emp_no, dept_no
   FROM dept_manager
   WHERE to_date = '9999-01-01'
),
manager_salaries AS (
   SELECT cm.emp_no, cm.dept_no, s.salary
   FROM current_managers cm
   JOIN salaries s
   ON cm.emp_no = s.emp_no
   WHERE s.to_date = '9999-01-01'
)
SELECT ms.emp_no, d.dept_name, ms.salary
FROM manager_salaries ms
JOIN departments d
```

```
ON ms.dept_no = d.dept_no
WHERE ms.salary = (SELECT MAX(salary) FROM manager_salaries);
```

Output:

	emp_no	dept_name	salary
▶	110039	Marketing	106491

8.Find which current department manager has less salary than highest salaried employee of that department.

Query:

```
WITH dept_max_salary AS (
   SELECT de.dept_no, MAX(s.salary) AS max_salary
   FROM dept_emp de
   JOIN salaries s
   ON de.emp_no = s.emp_no
   WHERE de.to_date = '9999-01-01'
     AND s.to_date = '9999-01-01'
   GROUP BY de.dept_no
),
manager_salary AS (
   SELECT dm.emp_no, dm.dept_no, s.salary
   FROM dept_manager dm
   JOIN salaries s
   ON dm.emp_no = s.emp_no
   WHERE dm.to_date = '9999-01-01'
     AND s.to_date = '9999-01-01'
)
SELECT
   m.emp_no,
   d.dept_name,
   m.salary,
   dms.max_salary
```

```
FROM manager_salary m

JOIN dept_max_salary dms

ON m.dept_no = dms.dept_no

JOIN departments d

ON m.dept_no = d.dept_no

WHERE m.salary < dms.max_salary;
```

Output:

emp_no	dept_name	salary	max_salary
110039	Marketing	106491	143644
110114	Finance	83457	134662
110228	Human Resources	65400	123674
110420	Production	56654	132552
110567	Development	74510	144434
110854	Quality Management	72876	121899
111133	Sales	101987	158220
111534	Research	79393	124181
111939	Customer Service	58745	144866

9.Which department manager has the highest salary among all departments. output : emp_no ,salary,dept_name

Query:

```
WITH manager_salaries AS (

   SELECT dm.emp_no, dm.dept_no, s.salary

   FROM dept_manager dm

   JOIN salaries s

   ON dm.emp_no = s.emp_no

   WHERE dm.to_date = '9999-01-01'

     AND s.to_date = '9999-01-01'

)

SELECT

   ms.emp_no,

   ms.salary,

   d.dept_name
```

```
FROM manager_salaries ms

JOIN departments d

ON ms.dept_no = d.dept_no

WHERE ms.salary = (SELECT MAX(salary) FROM manager_salaries);
```

Output:

	emp_no	salary	dept_name
▶	110039	106491	Marketing